

# Air Quality

## Pupil sheet

Name: \_\_\_\_\_

Date: \_\_\_\_\_

**Safety:** Do not move the boxes unless told to; the teacher opens the window; never climb on the sill.

### 1. What I see on the screen

The screen shows **one gauge per box** and a summary line:  $\Delta\text{CO}_2$ , the difference between the highest box and the lowest. With only one box it reads **Connect at least two boxes**. Each box has its own name and colour, the same on the gauge and on the chart.

**Under the  $\Delta$  line the app writes two lines.** The **Level** comes from the highest value: from 1200 ppm “the air is getting stale”, from 2000 ppm “open the windows now”. The **Difference** uses *fixed thresholds* — 300 and 600 ppm — which know nothing about our boxes. **I check it against my own threshold.**

My detection threshold (from co-location) = \_\_\_\_\_ ppm    A difference smaller than that *cannot be told from*

### 2. What I do on my own

**1. Co-location.** Every box on the same table, ten minutes. I record  $\Delta\text{CO}_2$  over the last five minutes; the largest value is the threshold above.

| Minute         | 6 | 7 | 8 | 9 | 10 |
|----------------|---|---|---|---|----|
| $\Delta$ (ppm) |   |   |   |   |    |

**First I predict.** The corner furthest from the window will be about \_\_\_\_\_ ppm higher than the spot beside it.

because \_\_\_\_\_

**2. Every location, at the same moment.** Time: \_\_\_\_\_ How many of us: \_\_\_\_\_ Window:

☐ shut ☐ open

| Location           | Box | CO <sub>2</sub> (ppm) | Band (level) |
|--------------------|-----|-----------------------|--------------|
| Far corner         |     |                       |              |
| By the window      |     |                       |              |
| By the door        |     |                       |              |
| Middle of the room |     |                       |              |

**3. I compare every pair with my threshold. “Borderline” is a correct answer, not a dodge.**

| The pair             | $\Delta$ (ppm) | Verdict (circle one)                 |
|----------------------|----------------|--------------------------------------|
| Far corner vs window |                | detected / borderline / not detected |
| Far corner vs door   |                | detected / borderline / not detected |
| Middle vs window     |                | detected / borderline / not detected |

**4. Airing.** We open the window.  $\Delta$ CO<sub>2</sub> immediately afterwards: \_\_\_\_\_ ppm. Did it rise?

☐ yes ☐ no

After ten minutes: \_\_\_\_\_ ppm.

### 3. The conclusion I write

I predicted that, and the data show that \_\_\_\_\_

When we opened the window,  $\Delta$  rose because \_\_\_\_\_

The location most in need of air is \_\_\_\_\_, because \_\_\_\_\_

#### Exit ticket

What is our detection threshold, and how do we know it?